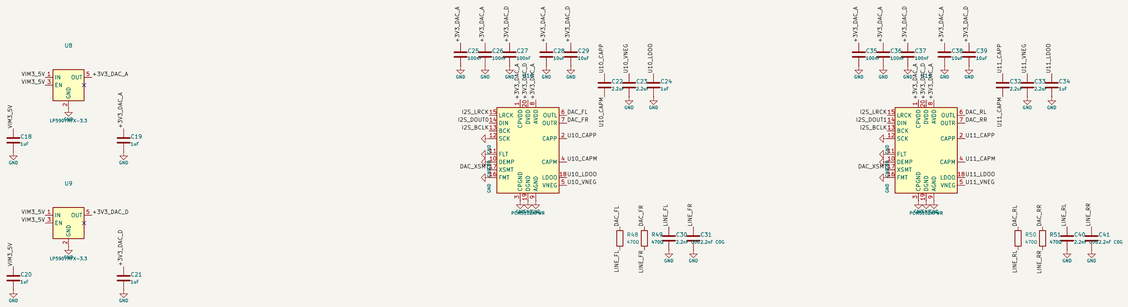
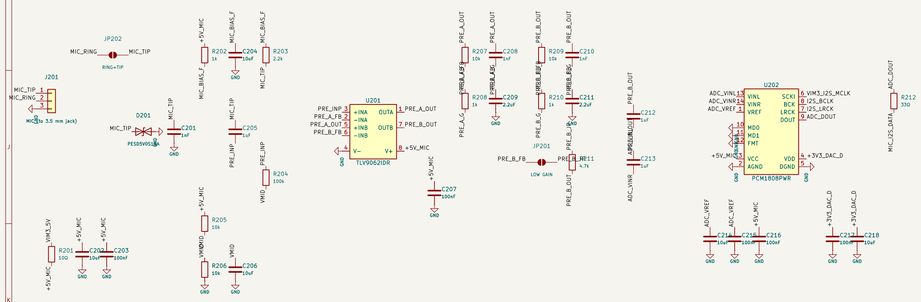
AUDIO MODULE rev 0.7: I2S from J24 -> 2x PCM5102A (front + rear) -> line out back through J24 to ISO C1 on the main board



2.1 Vrms ground-centred line out, no coupling caps, 24-bit in 32-bit I2S slots, 48/96/192 kHz.

CABIN MIC: 3.5 mm stereo panel jack -> J201 (JST XH) -> electret bias -> 2-stage preamp (TLV9062) -> PCM1808 ADC -> J24 pin 11 (VIM3 pin 35)



Gain: about +42 dBV/Pa car electret, speech at the A-pillar about 65 dB SPL -> about -30 dBFS at 42 dB. Raise or lower further in the HAL.

Car radio audio module rev 0.7 — plugs onto the peripheral board (J24 -> J23). Generated by gen_audio7.py.

Sheet: /	
File: carradio_audio_rev07.kicad_sch	
Title: Car radio audio module (2x PCM5102A + mic ADC)	
Size: A1	Date: 2026-11-02
KiCad E.D.A. 10.0.0	Rev: 0.7
	Id: 1/1